

336K

Total Inventory

\$4.51M

Total Production Quantity

11K

Total Scrap Quantity

\$63.79M

Total Purchase Cost

4012

Total Purchase Orders

104

Total Vendors

OrderDate, Year, Month, Day

All

Product category

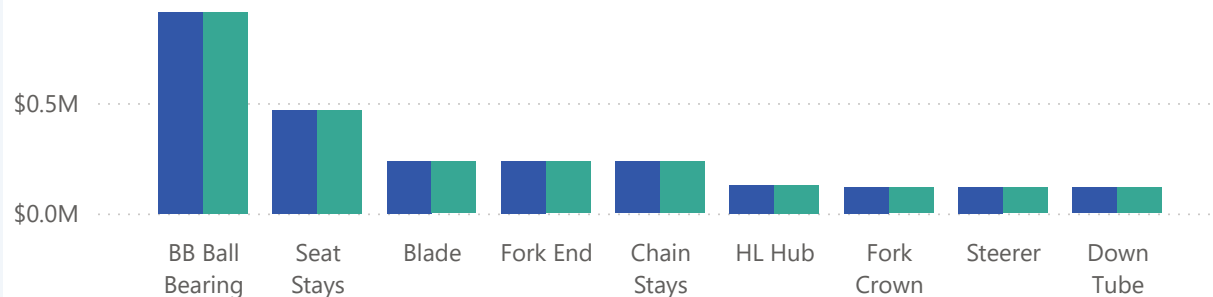
All

Product

All

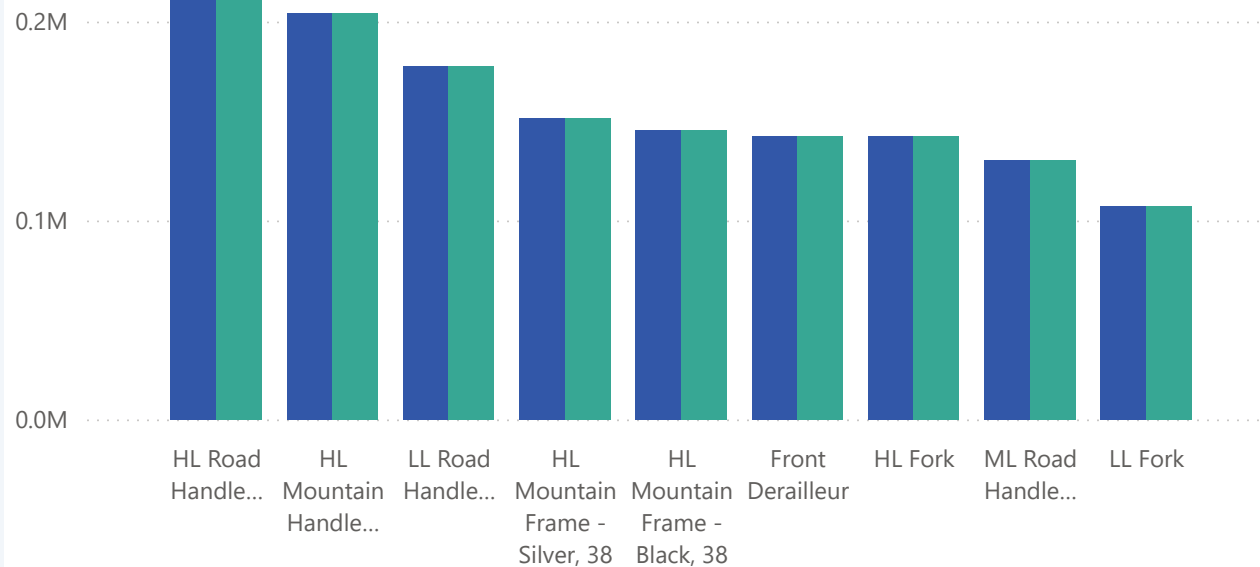
Total Production Quantity and Total Stocked Quantity by Product

● Total Production Quantity ● Total Stocked Quantity

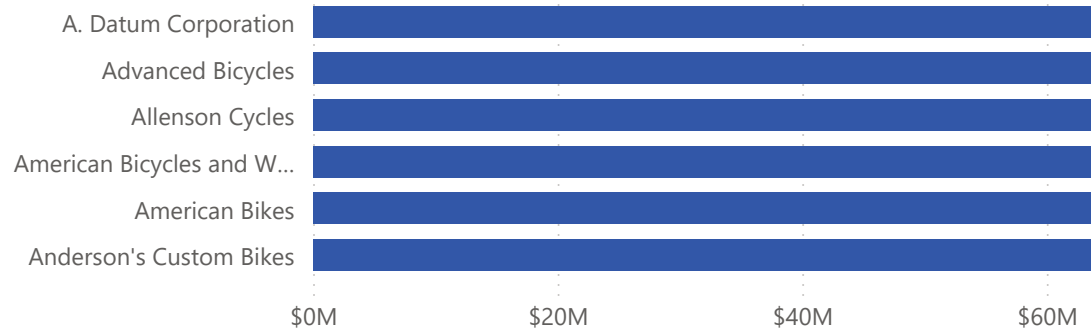


Total Planned Production Cost and Total Actual Production Cost by Product

● Total Planned Production Cost ● Total Actual Production Cost



Total Purchase Cost by Vendor



336K

Total Inventory

432

Products With Inventory

14

Inventory Locations

100.00%

Inventory Turnover Rate

777.72

Average Inventory Per Product

Product Category

All

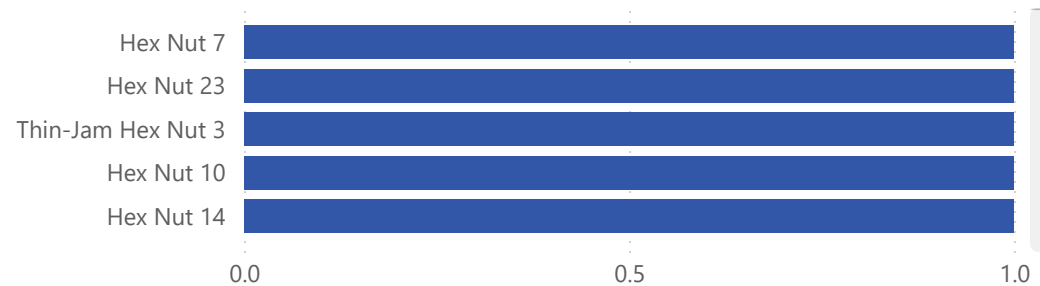
Product Subcategory

All

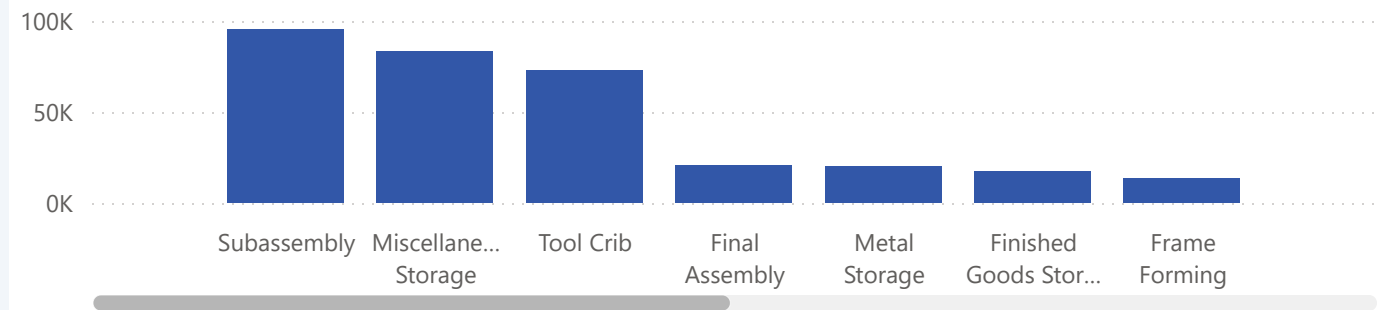
Location

All

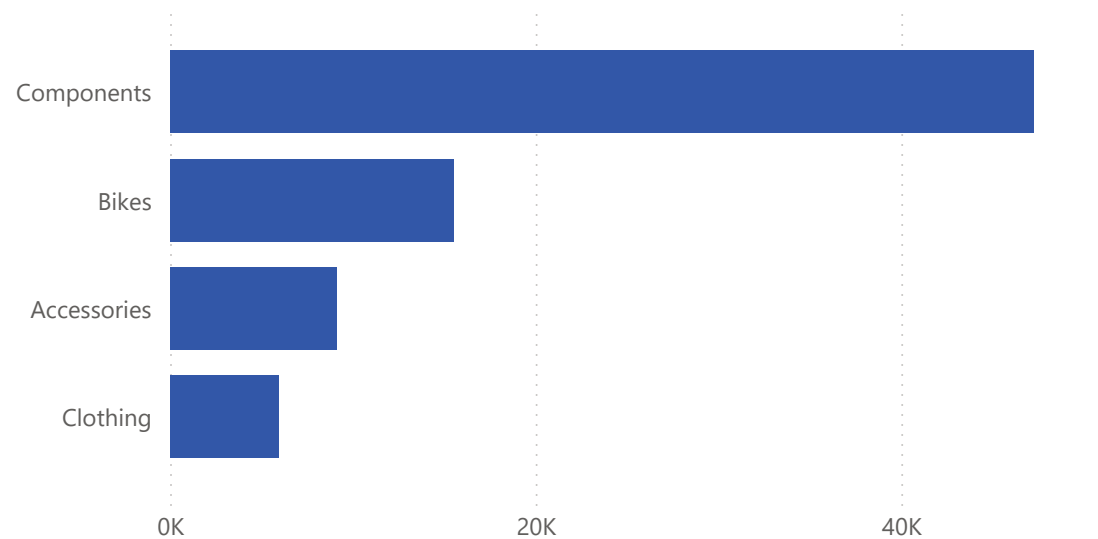
Top 5 products by Inventory



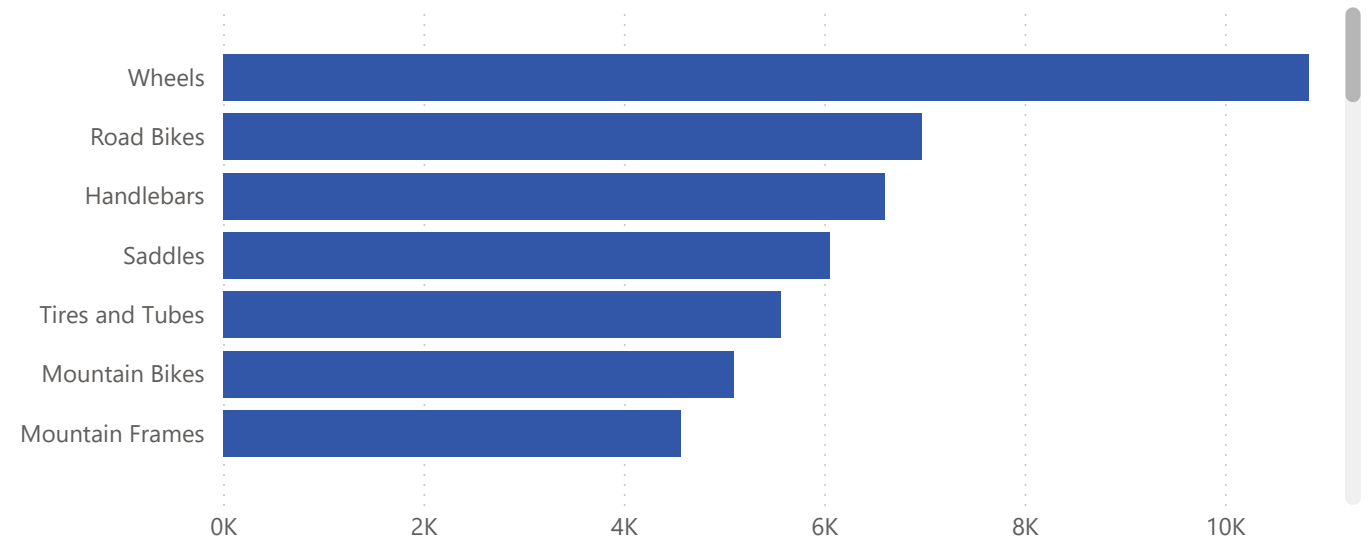
Total Inventory by Location



Total Inventory by Category



Total Inventory by Subcategory



\$4.51M

Total Production Quantity

4M

Total Stocked Quantity

11K

Total Scrap Quantity

0.24%

Scrap Rate %

0.00%

Production Cost Variance %

Product Category

All

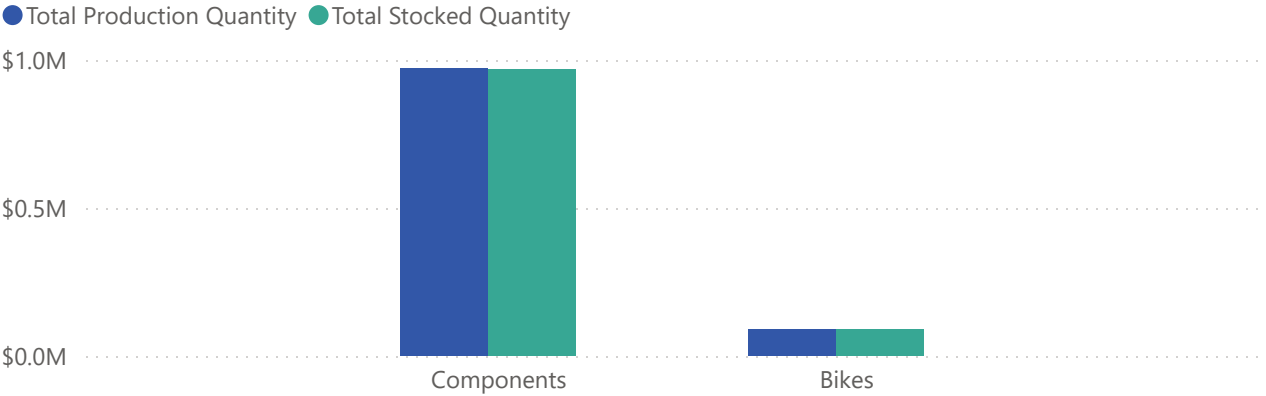
Product Subcategory

All

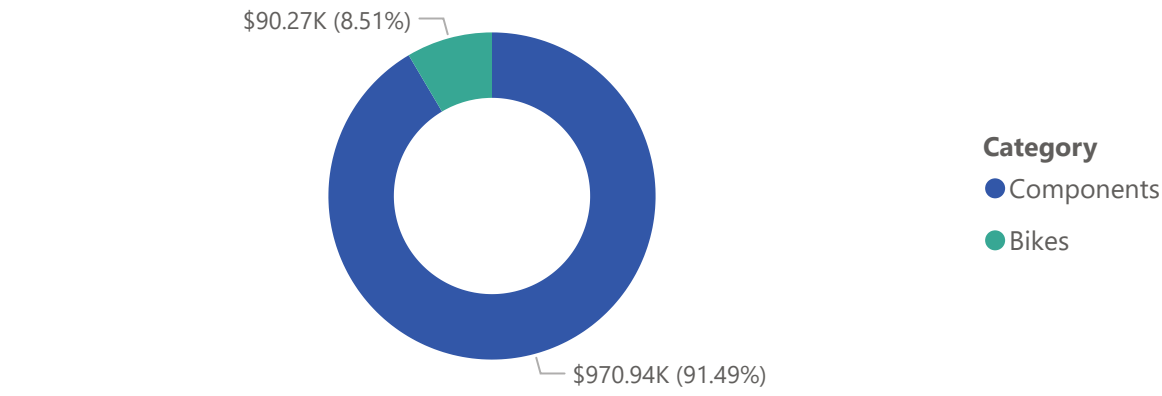
Product

All

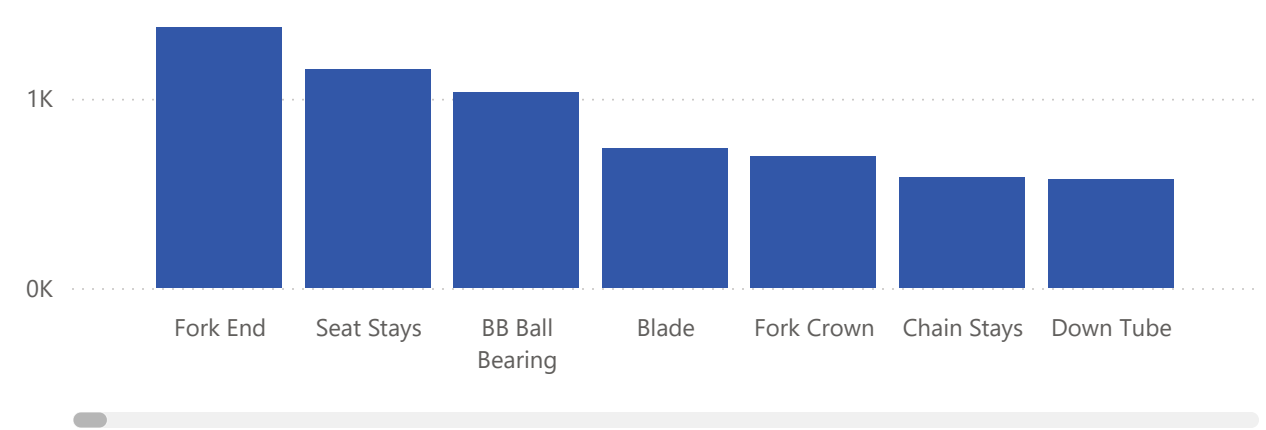
Total Production Quantity and Total Stocked Quantity by Product



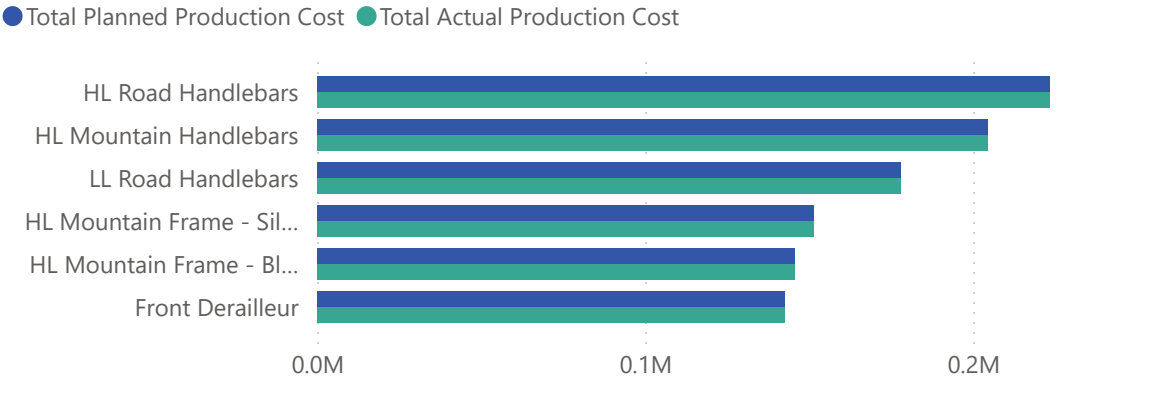
Production Quantity by Category



Total Scrap Quantity by Product



Total Planned Production Cost and Total Actual Production Cost by Product Name



\$63.79M

Total Purchase Cost

4012

Total Purchase Orders

104

Total Vendors

2M

Total Quantitv Purchased

3.12%

Vendor Defect Rate %

15.90K

Average Purchase Order Value

Product

All

Product Category

All

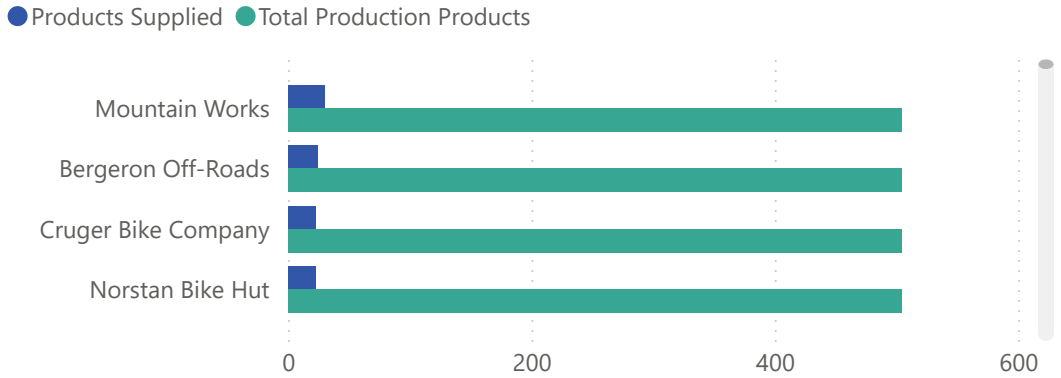
Shipping Method

All

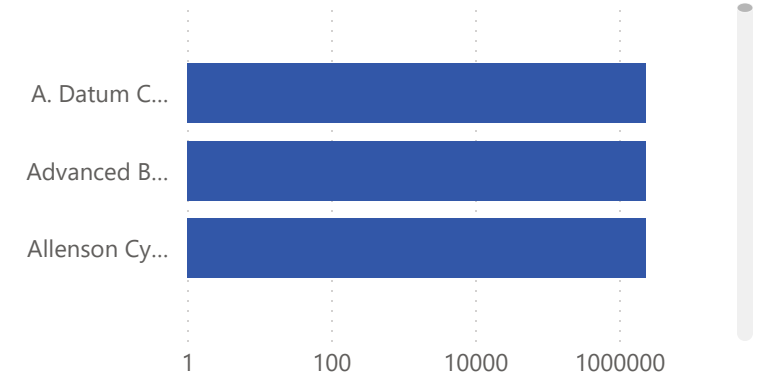
Vendor

All

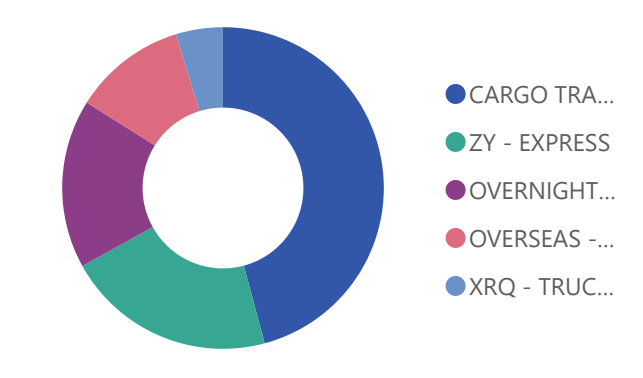
Products Supplied and Total Production Products by Vendor



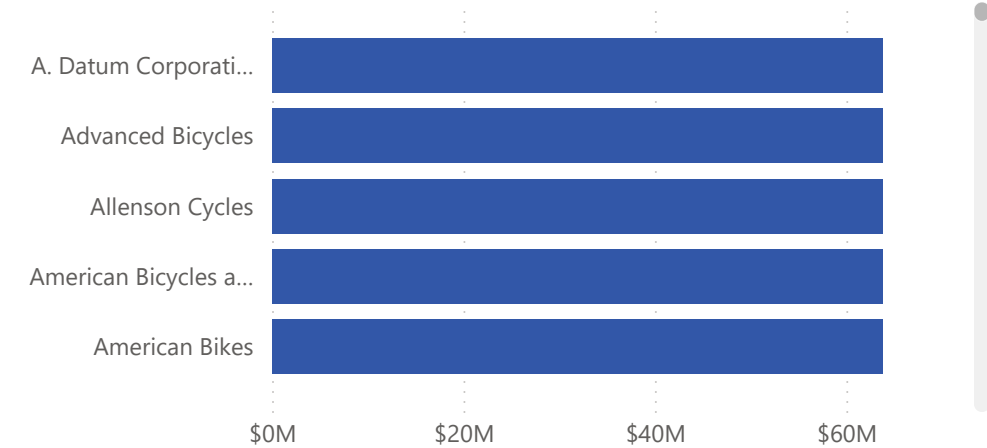
Total Quantity Purchased by Vendor



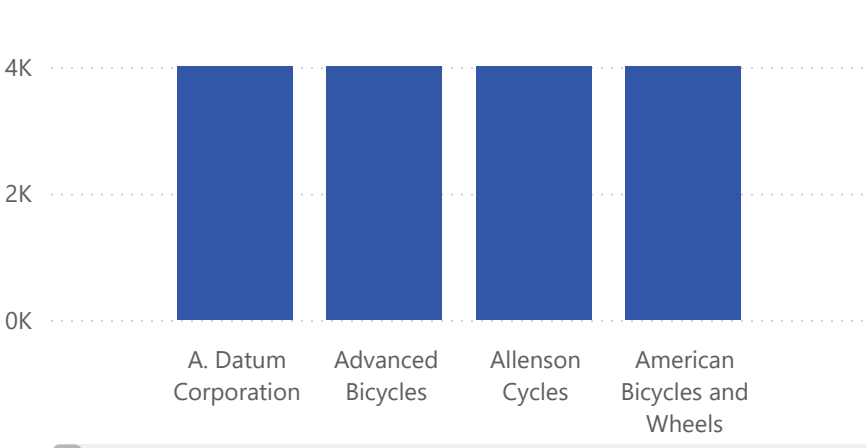
Total Purchase Cost by Shipping Method



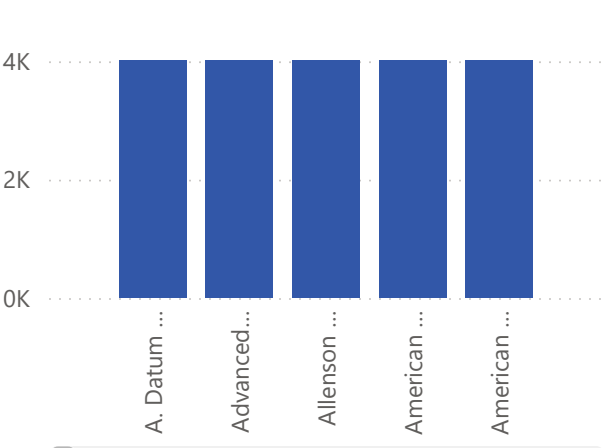
Total Purchase Cost by Vendor



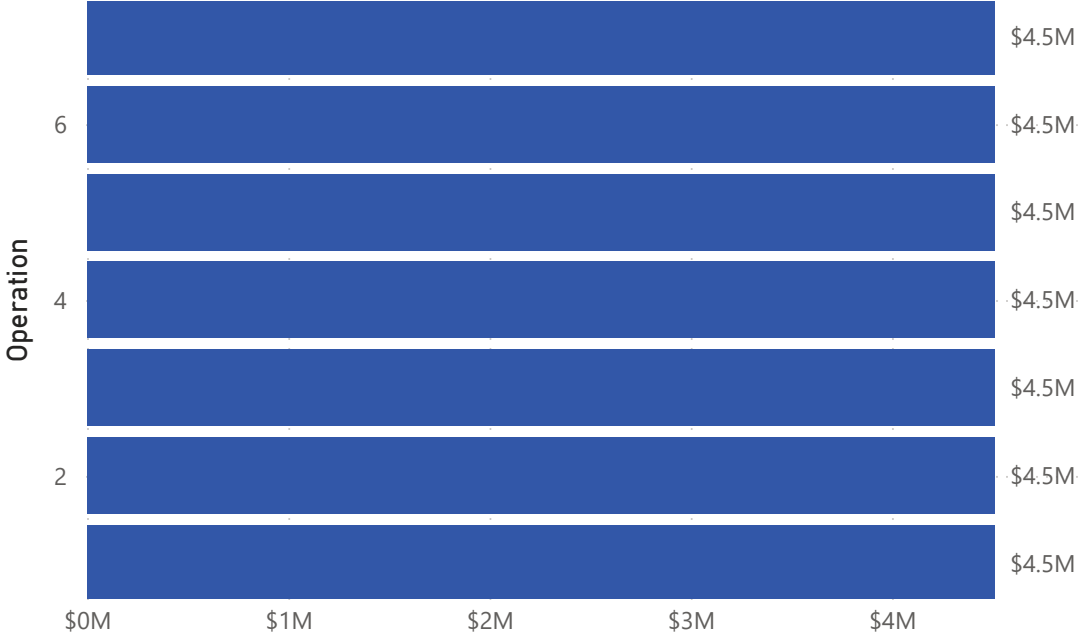
Total Purchase Orders by Vendor



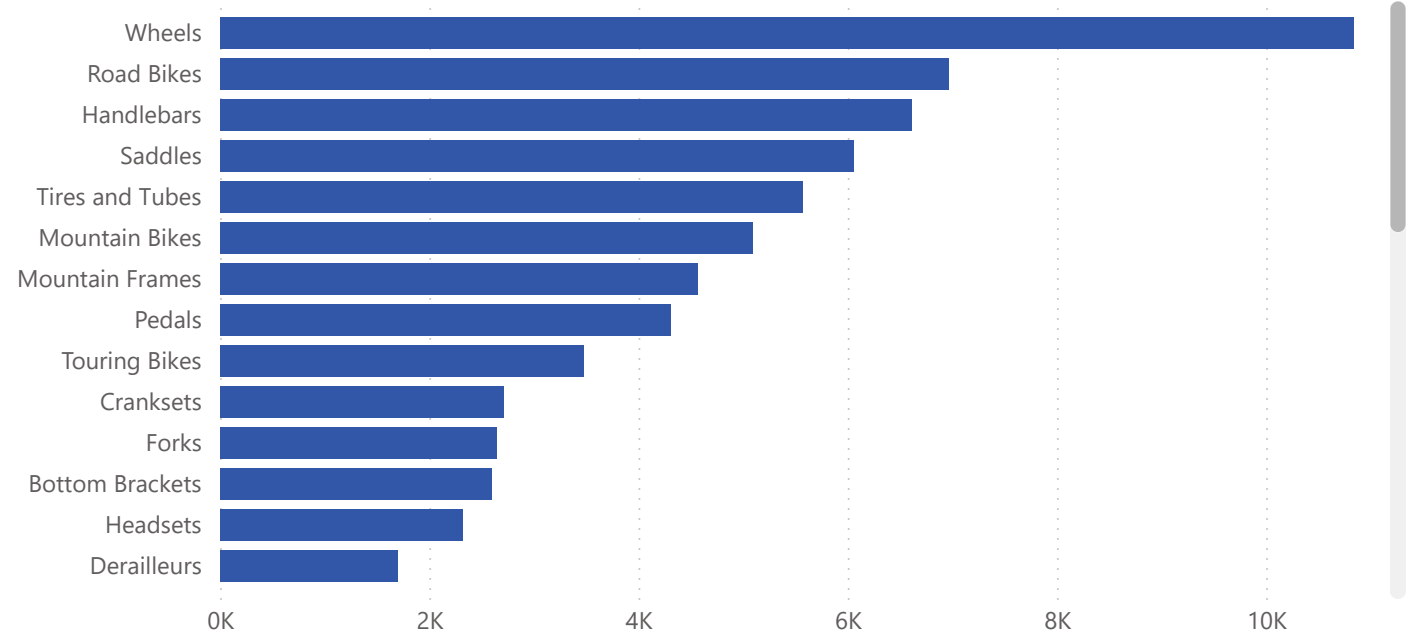
Rejected Order Count by Vendor



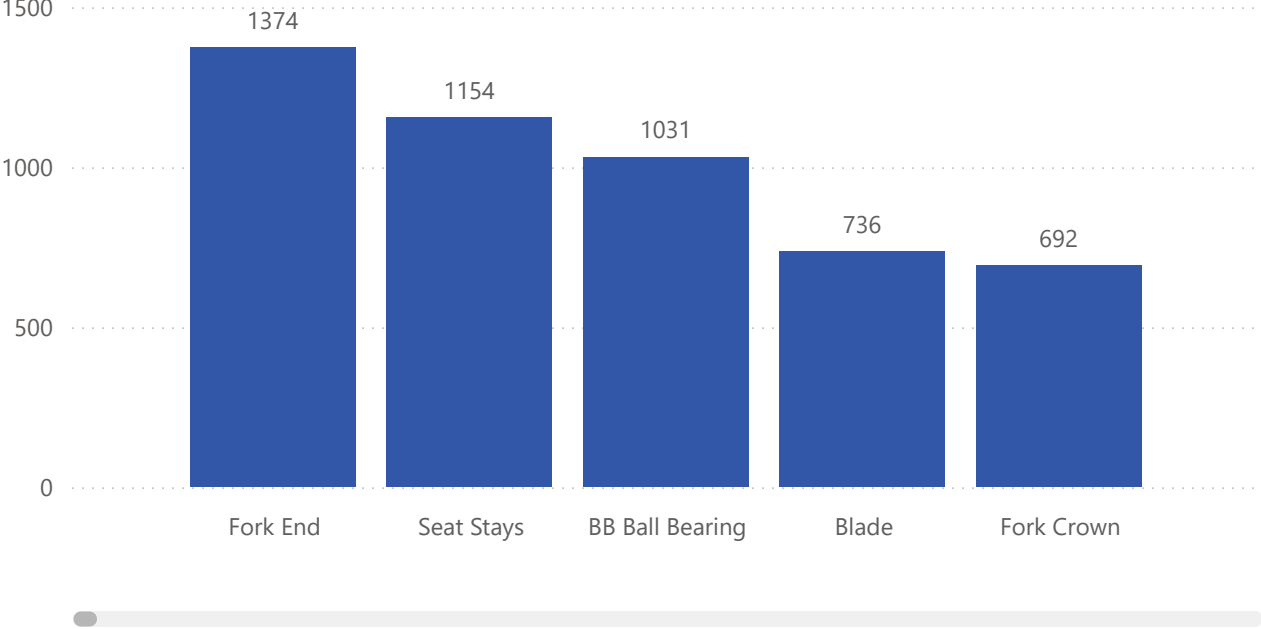
Production Quantity by Operation



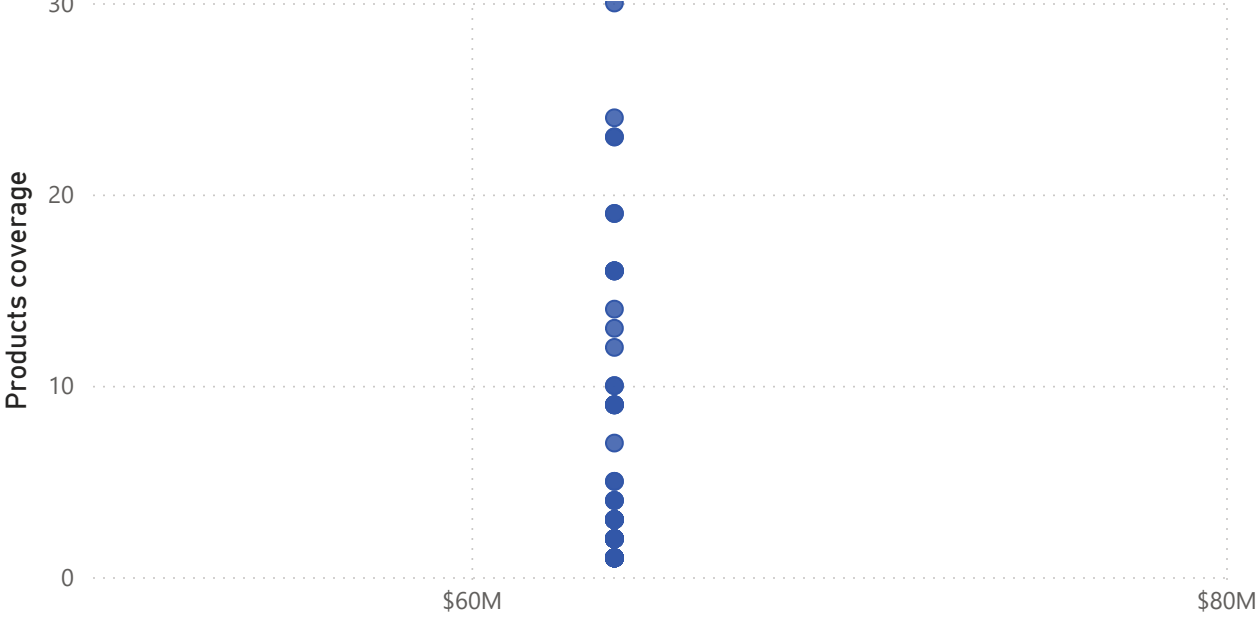
Total Inventory by Product Subcategory



Scrap Quantity by Product



Vendor spend and Products coverage by Product



ADVENTUREWORKS EXECUTIVE SUMMARY REPORT

Scope: Manufacturing Performance, Sourcing Strategy & Inventory Management

Prepared For: Stake Holders, Operational Leadership & Department Managers

Prepared By: Eke Ifeomachukwuyerem

CURRENT OPERATIONAL PERFORMANCE

- **Total Purchase Cost: \$63.79M** invested across 4,012 completed purchase orders.
- **Total Production Quantity: 4.51M total units produced**, with **4.0M units transferred to active inventory**.
- **Total Inventory: 336K units on hand** distributed across 14 operational fulfillment centers.
- **Total Vendors: 104 active vendors** supporting component sourcing and material supply.
- **Scrap Rate: 0.24%** (11K discarded units), demonstrating strong internal manufacturing quality control.
- **Vendor Defect Rate: 3.12%** across incoming shipments, representing a key area for operational improvement.

Operational Drivers & Key Findings

- **Strict Budget Compliance:** Planned production costs matched actual execution costs (0.00% variance), showing strong cost control across major product lines.
- **Component-Driven Volume:** Output is heavily concentrated in component fabrication (**91.5% / 970.94K units**), while finished bike assembly accounts for **8.5% (90.27K units)**.
- **Scrap Quality:** Factory waste is tightly contained overall (0.24%), but discarded material is heavily concentrated at three steps: **Fork Ends, Seat Stays, and Ball Bearings**.
- **Vendor Risk:** Sourcing spend is disproportionately absorbed by top vendors (A. Datum Corp, Advanced Bicycles), despite having 104 active sellers.
- **Incoming Quality Bottlenecks:** A **3.12% vendor defect rate** creates re-inspection overhead and slows down material feed lines.
- **Inventory Capital Allocation:** Working capital is tied up predominantly in **Subassembly Storage**. Wheels, Road Bikes, and Handlebars occupy the largest physical floor space.

BUSINESS IMPACT & ACTION PLAN

Business Impact & Strategic Risk

1. **Vendor Defect Friction:** A hours on re-inspections. Capping defects below 1.5% protects 3.12% defect rate wastes worker assembly lines.
2. **Capital Tie-Up:** Holding 336K units primarily in subassemblies limits cash flow. Aligning component stock closer to immediate demand frees tied-up capital.
3. **Single-Source Dependency:** Concentrating \$63.79M across top vendors leaves operations vulnerable to supplier production delays.

Strategic Management Initiatives

- **Initiative 1 (Reallocate Spend & Enforce Service Level Agreement):** Shift 15%–20% of orders to secondary vendors among our 104 suppliers. Add contract penalty clauses capping vendor defects at 1.5%.
- **Initiative 2 (Target Quality Audits on Scrap Lines):** Recalibrate tooling on top 3 scrap stations (**Fork Ends, Seat Stays, Ball Bearings**) to cut scrap from 11K to under 5K units.
- **Initiative 3 (Rebalance Subassembly Stock Triggers):** Adjust buffer stock levels in subassembly storage to transition toward demand-driven inventory control across all 14 locations.
- **Initiative 4 (Diversify Shipping Options):** Contract secondary regional carriers alongside primary cargo transport to handle rush orders and avoid shipping bottlenecks.